Table 1: Notation and definitions

Notation	Meaning
$\mathbb{N}$	The natural numbers $\{1, 2, 3, \dots\}$.
$\mathbb{R}$	The real numbers.
$\mathbb{R}_{>0}$	The positive real numbers $\{x \in \mathbb{R} : x > 0\}$.
$\mathbb{R}_{\geq x}$	The set $\{y \in \mathbb{R} : y \geq x\}$ for a given $x \in \mathbb{R}$.
$\#S$	The cardinality of a set S.
$A \times B$	The Cartesian product $\{(a, b) : a \in A,\, b \in B\}$.
$\mathbb{R}^d$	d-dimensional real space.
$A \subseteq B$	A is a subset of B, possibly $A = B$.
$A \subsetneq B$	A is a proper subset of B, i.e. $A \subseteq B$ and $A \neq B$.
$\mathbb{I}\{x \in S\}$	Indicator of the set S, equal to 1 if $x \in S$ and 0 otherwise.
$\mathrm{int}(S)$	The interior of $S \subseteq \mathbb{R}$ under the usual Euclidean topology.
$\phi(A)$	For $A \subseteq X$, the image under $\phi : X \to Y$ is $\phi(A) := \{\phi(a) : a \in A\}$.
$\phi^{-1}(B)$	For $B \subseteq Y$, the pre-image under $\phi : X \to Y$ is $\phi^{-1}(B) := \{x \in X : \phi(x) \in B\}$.
$\log(\cdot)$	The natural logarithm $\log : (0, \infty) \to \mathbb{R}$.
$\phi \propto \tilde{\phi}$	Functions $\phi, \tilde{\phi} : D \to \mathbb{R}$ are proportional if $\exists c \in \mathbb{R}$ such that $\phi(x) = c\,\tilde{\phi}(x)$ for all $x \in D$.
$\phi_2 \circ \phi_1$	For $\phi_1, \phi_2 : \mathbb{R} \to \mathbb{R}$, the composition satisfies $(\phi_2 \circ \phi_1)(x) = \phi_2(\phi_1(x))$ for all $x \in \mathbb{R}$.
$\mathbf{x}, \mathbf{X}$	Vectors in $\mathbb{R}^n$, where $n \in \mathbb{N}$.
$\mathbf{A}$	Matrix in $\mathbb{R}^{n \times m}$, where $m, n \in \mathbb{N}$.
$\mathbf{A}^T$	The transpose of a matrix $\mathbf{A} \in \mathbb{R}^{n \times m}$.
X	Real-valued random variable $X : \Omega \to \mathbb{R}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with realization x.
$\mathbf{X}$	n-dimensional random vector $\mathbf{X} : \Omega \to \mathbb{R}^n$ on $(\Omega, \mathcal{F}, \mathbb{P})$, with realization $\mathbf{x}$.
f_X	The probability density function (PDF) of X.
$f_{\mathbf{X}}$	The joint PDF of $\mathbf{X}$.
F_X	The cumulative distribution function (CDF) of X.
$F_{\mathbf{X}}$	The joint CDF of $\mathbf{X}$.
$\mathbb{E}[g(X)]$	Expectation of $g(X)$, typically $\mathbb{E}[g(X)] = \int g(x)\, f_X(x)\, dx$ when X has density f_X.
$\mathbb{E}[g(\mathbf{X})]$	Expectation of $g(\mathbf{X})$, typically $\mathbb{E}[g(\mathbf{X})] = \int g(\mathbf{x})\, f_{\mathbf{X}}(\mathbf{x})\, d\mathbf{x}$ when $\mathbf{X}$ has joint density $f_{\mathbf{X}}$.

Table 2: Probability notation and distributions

Notation	Meaning
$\mathrm{Var}[g(X)]$	Variance of the random variable $g(X)$, typically $\mathrm{Var}[g(X)] = \mathbb{E}\big[(g(X) - \mathbb{E}[g(X)])^2\big]$.
$\mathrm{Var}[g(\mathbf{X})]$	Variance of the random variable $g(\mathbf{X})$, where $g : \mathbb{R}^n \to \mathbb{R}$.
$\mathrm{Supp}(f)$	Support of $f : X \subseteq \mathbb{R}^n \to \mathbb{R}$, i.e. $\mathrm{Supp}(f) = \{\mathbf{x} \in X : f(\mathbf{x}) \neq 0\}$.
$X \sim f_X$	The random variable X has PDF f_X.
$X \overset{d}{=} Y$	Random variables X and Y are equal in distribution (have the same law).
$X_n \xrightarrow{a.s.} X$	Sequence $(X_n)_{n\geq 1}$ converges almost surely to X if $\mathbb{P}\big(\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}\big) = 1$.
$X_n \xrightarrow{d} X$	Sequence $(X_n)_{n\geq 1}$ converges in distribution to X if $F_{X_n}(x) \to F_X(x)$ at all continuity points x of F_X.
$F_n(x) \xrightarrow{d} F(x)$	The CDFs F_n converge in distribution to F if $F_n(x) \to F(x)$ at all continuity points of F.
$\mathrm{Bern}(p)$	Bernoulli distribution with success probability $p \in [0,1]$. The PMF $f : \{0,1\} \to [0,1]$ is $f(x) = p\,\mathbb{I}\{x = 1\} + (1-p)\,\mathbb{I}\{x = 0\}$.
$\mathrm{Unif}(a,b)$	Continuous uniform distribution on $[a,b] \subset \mathbb{R}$, $a < b$. The PDF $f : \mathbb{R} \to [0, \frac{1}{b-a}]$ is $f(x) = \frac{1}{b-a}\,\mathbb{I}\{x \in [a,b]\}$.
$\mathcal{N}(\mu, \sigma^2)$	Normal (Gaussian) distribution with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 \in \mathbb{R}_{>0}$. The PDF $f : \mathbb{R} \to \mathbb{R}_{>0}$ is $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big\{-\frac{(x-\mu)^2}{2\sigma^2}\Big\}$.
$\mathcal{T}f_X(\boldsymbol{\theta}, a)$	Truncation of a PDF f_X at threshold $a \in \mathbb{R}$ (parameters $\boldsymbol{\theta} \in \mathbb{R}^k$). The truncated PDF $\mathcal{T}f_X : \mathbb{R}_{\geq a} \to \mathbb{R}$ is $\mathcal{T}f_X(x) = \big[\int_a^\infty f_X(t)\,dt\big]^{-1}\mathbb{I}\{x \geq a\} f_X(x)$.
$\mathrm{Cauchy}(x_0, \gamma)$	Cauchy distribution with location $x_0 \in \mathbb{R}$ and scale $\gamma \in \mathbb{R}_{>0}$. The PDF $f : \mathbb{R} \to \mathbb{R}_{>0}$ is $f(x) = \frac{1}{\pi\gamma}\left[1 + \left(\frac{x-x_0}{\gamma}\right)^2\right]^{-1}$.
$\mathrm{Expo}(\lambda)$	Exponential distribution with rate $\lambda \in \mathbb{R}_{>0}$. The PDF $f : \mathbb{R} \to \mathbb{R}_{>0}$ is $f(x) = \lambda e^{-\lambda x}\,\mathbb{I}\{x \geq 0\}$.
$\mathrm{Beta}(\alpha, \beta)$	Beta distribution with shape parameters $\alpha, \beta \in \mathbb{R}_{>0}$. The PDF $f : (0,1) \to \mathbb{R}_{>0}$ is $f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}\,\mathbb{I}\{0 < x < 1\}$.
$\mathrm{Gamma}(\alpha, \beta)$	Gamma distribution with shape $\alpha \in \mathbb{R}_{>0}$ and rate $\beta \in \mathbb{R}_{>0}$. The PDF $f : \mathbb{R} \to \mathbb{R}_{>0}$ is $f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}\,\mathbb{I}\{x > 0\}$.
$\mathrm{Weib}(\alpha, \beta)$	Weibull distribution with shape $\alpha \in \mathbb{R}_{>0}$ and scale $\beta \in \mathbb{R}_{>0}$. The PDF $f : \mathbb{R} \to \mathbb{R}_{>0}$ is $f(x) = \frac{\alpha}{\beta^\alpha} x^{\alpha-1} \exp\big\{-(x/\beta)^\alpha\big\}\,\mathbb{I}\{x \geq 0\}$.